GOLDEN ARROWS DATASET V3.1

Simulation-Generated Multi-Drone Swarm Transition Dataset

Dataset Description | Version 3.1

DATASET AT A GLANCE

1,000 MISSIONS	**113,954** TRANSITIONS
3,418,620 DRONE-TRANSITION RECORDS	**30** DRONES / TRANSITION

NOTE — RECORD COUNT INTERPRETATION
The 3,418,620 drone-transition records are derived from 113,954 transitions × 30 drones per transition. These records are nested within transitions and missions and should not be treated as 3,418,620 independent samples.

1. Abstract

Golden Arrows Dataset V3.1 is a simulation-generated dataset of multi-drone swarm missions with rich transition-level records. It contains 1,000 missions across PATROL, SURVEILLANCE, SEARCH, and RECOVERY, with 250 missions per type. Advanced Swarm Behavior is enabled throughout. The package provides raw mission JSON, processed transition JSONL, metadata, schemas, optional preprocessing examples, and validation materials.

The transition representation preserves mission, decision, action, result, network, swarm, and drone-level information for descriptive and methodological research. This is simulated data, not real-world flight telemetry. It makes no claim about real-world safety, operational effectiveness, deployment readiness, or real-world performance.

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2. Dataset Identity and Motivation

V3.1 is a separate release that expands V3.0 from 100 to 1,000 missions while preserving rich transition- and drone-level information. V3.0 remains frozen and unchanged. V3.1 supports descriptive and methodological research; it does not establish comparative mission performance or real-world equivalence.

Property	Verified value
Dataset / version	Golden Arrows Dataset V3.1 / 3.1
Dataset type	Simulation-generated multi-drone swarm transition dataset
Missions / transitions	1,000 / 113,954
Drone-transition records	3,418,620 (derived)
Drones per transition	30
Mission types	PATROL, SURVEILLANCE, SEARCH, RECOVERY
Advanced Swarm Behavior	Enabled throughout
Time step / maximum steps	dt = 0.5 / 1,000
Collection seed	20260916

3. Dataset Composition

The mission-family counts are balanced; the transition totals differ because the recorded missions contain different numbers of transitions.

Mission type	Missions	Transitions
PATROL	250	63,694
SURVEILLANCE	250	22,036
SEARCH	250	11,856
RECOVERY	250	16,368
Total	1,000	113,954

The four mission labels identify the configured mission families. Balanced mission counts do not imply comparative effectiveness, difficulty, or real-world equivalence.

Transition counts are not mission counts. Each transition contains 30 drone records; the resulting 3,418,620 drone-transition records are nested within transitions and missions and are not independent samples.

How to read the totals

The published totals describe the V3.1 simulation collection. Mission distribution counts distinct missions; transition counts describe the rich transition records in those missions. The drone-transition record total is a derived cardinality, not a separate count of independent observations.

4. Package Organization

The published package contains the files shown below. Raw mission JSON and processed transition JSONL are stored under the indicated data directories; the data directories contain 1,000 files each.

```
Golden_Arrows_Dataset_V3.1/
|-- README.md
|-- LICENSE
|-- CITATION.cff
|-- dataset_description.pdf
|-- dataset_card.md
|-- CHANGELOG.md
|-- data/
|   |-- metadata.csv
|   |-- raw/missions/             [1,000 JSON files]
|   `-- processed/transitions/    [1,000 JSONL files]
|-- schema/
|   |-- mission_schema.json
|   |-- transition_schema.json
|   `-- behavior_schema.json
|-- validation/
|   |-- dataset_validation_report.txt
|   `-- dataset_statistics.json
`-- examples/preprocessing/
    |-- README.md
    |-- prepare_state_only.py
    `-- prepare_behavior_aware.py
```

5. Simulation and Mission Context

The dataset records a simulated swarm of 30 drones across four configured mission types. PATROL represents patrol-oriented missions; SURVEILLANCE represents monitoring-oriented missions; SEARCH represents search-oriented missions; and RECOVERY represents recovery-oriented missions. These are descriptive labels for the configured mission families, not rankings or claims of comparative difficulty.

Raw mission records preserve mission context. Processed JSONL records represent individual transitions and include mission, transition, decision, action, result, network, and swarm sections. Consult the included schema materials for field definitions.

6. Data Generation and Reproducibility

V3.1 documents one simulation collection configuration. Reproducibility materials include mission metadata, raw and processed data, schemas, validation artifacts, and deterministic optional preprocessing examples.

Configuration	Verified value
Drones / missions	30 / 1,000 (250 per mission type)
Time step / maximum steps	dt = 0.5 / max_steps = 1,000
Collection seed	20260916
Mission seed range	20260917 through 20261916
Dataset baseline	V2.2_FROZEN (generation-baseline reference only)
Public dataset identity	Golden Arrows Dataset V3.1
Policy / behavior mode	SCRIPTED_MISSION_CONTROLLER+ADVANCED_SWARM_BEHAVIOR / BOUNDED_CORRECTION_ON_MISSION_MOTION

V2.2_FROZEN identifies a generation-baseline reference; it is not the identity of the published V3.1 dataset.

7. Conceptual System Architecture

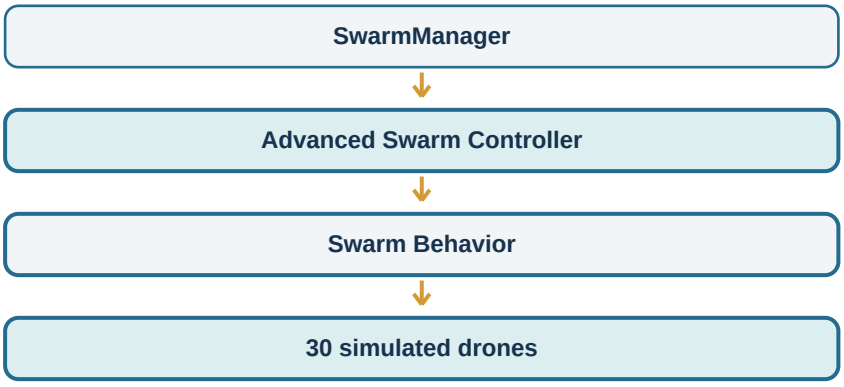


Figure 1. Conceptual architecture terminology documented for V3.1.

8. Transition Structure

A transition captures the progression STATE -> ACTION -> RESULT -> NEXT STATE. The verified top-level sections are schema_version, mission, transition, decision, action, result, network, and swarm. Each transition contains records for 30 drones.

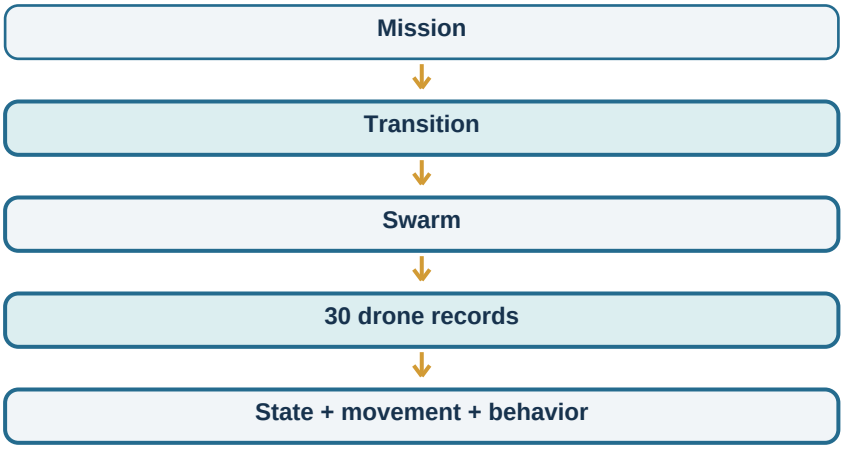


Figure 2. Conceptual organization of a rich transition; not an implementation schematic.

9. Advanced Swarm Behavior

Advanced Swarm Behavior telemetry is enabled throughout V3.1. The documented components are descriptive names for recorded behavior-related information; they are not asserted to be independent, learned, or validated in real-world flight, and they do not provide safety guarantees.

Component	Recorded description
Cohesion	Cohesion-related movement behavior
Alignment	Alignment-related movement behavior
Separation	Local separation behavior
Formation	Formation-related movement behavior
Target Attraction	Attraction toward the mission target
Network Preservation	Network/connectivity-preservation behavior
Boundary Avoidance	Movement away from boundary-risk conditions

10. State-Only Preprocessing Example

`examples/preprocessing/prepare_state_only.py` is an optional executable example. It reads the published rich transitions and can produce local tabular artifacts; those derivatives are not part of the publication package.

Category	Fields
Model inputs (13)	active_links; connected_drones; progress_before; position_x; position_y; altitude; speed; direction_x; direction_y; battery; signal_strength; latency; packet_loss
Metadata	mission_id; mission_type; drone_id; tick
Targets	next_position_x; next_position_y
Excluded	progress_after; actual_movement; behavior; action; decision

position_x/y use state_before.position; target next_position_x/y use the resulting state_after.position. Several other physical/network values are recorded under state_after. These fields are not automatically valid pre-action observations.

11. Behavior-Aware Preprocessing Example

`examples/preprocessing/prepare_behavior_aware.py` is optional and reads the published rich transitions. Its feature representation combines base state/context fields with recorded Advanced Swarm Behavior telemetry. It is a preprocessing example, not a trained policy, an RL action, or a performance claim.

Feature group	Count
Base state/context	13
Behavior-level telemetry	6
Behavior-vector features	8
Behavior-component features	28
Total model input features	55

Feature group	Recorded features
Behavior-level (6)	speed_scale; neighbor_count; minimum_neighbor_distance; formation_error; connectivity_risk; boundary_risk
Behavior vectors (8)	behavior_current_position_x/y; behavior_current_velocity_x/y; behavior_desired_velocity_x/y; behavior_movement_vector_x/y
Components (7 x 4 = 28)	cohesion; alignment; separation; formation; target_attraction; network_preservation; boundary_avoidance. Each records x, y, weight, magnitude.
Targets	next_position_x; next_position_y

The behavior vectors record x/y values for current position, current velocity, desired velocity, and movement vector. The component values describe recorded telemetry and should be interpreted in their transition context. The example keeps metadata separate and excludes actual_movement, action, decision, progress_after, and state_after from model inputs.

Targets are next_position_x and next_position_y. The presence of behavior telemetry does not make it independent of transition state or a learned behavior representation.

12. Temporal Semantics

Transition fields are recorded at different points in a transition, so field names and storage locations alone do not establish when a value was observable.

Field or group	Documented interpretation
state_before.position	Represents the current position used by the examples.
state_after physical/network values	Some physical and network values are stored here; do not assume they are pre-action.
Advanced Swarm Behavior	Telemetry is recorded separately from actual_movement.
next_position_x/y	Represent the resulting state_after.position.
action, decision, progress, network, behavior	Each has its own transition semantics; review timing for the task.

Fields stored under state_after must not automatically be interpreted as valid pre-action observations.

Researchers should inspect temporal ordering before using the records for supervised learning, causal analysis, reinforcement learning, multi-agent reinforcement learning, forecasting, or control-policy learning. The examples document particular feature choices; they do not resolve all timing questions for every research task.

13. Data Leakage and Feature Interpretation

Excluding selected fields from a feature list does not guarantee that all leakage has been eliminated. Independently review feature timing, target construction, mission structure, temporal dependencies, identifiers, and transition semantics. Choose a split and feature definition appropriate to the intended analysis.

Researcher timing review

Review point	Question to resolve for the task
Observation time	Which values are available before the modeled action, and which are recorded under state_after?
Target timing	How does the next-position target relate to the resulting state_after.position?
Behavior and control	How are behavior telemetry, actual_movement, action, and decision ordered and interpreted?
Dependence and split	Are related transitions and drone records kept together at the mission level?
Analysis design	Have identifiers, mission structure, and temporal dependencies been reviewed for the chosen task?

14. Mission-Level Splitting

The optional examples use a deterministic, stratified 68% / 16% / 16% split at the mission level with split seed 20260916. These are mission counts, not CSV row counts; generated split files are not included.

Split	Missions overall	Missions per type
Train (68%)	680	170
Validation (16%)	160	40
Test (16%)	160	40
Total	1,000	250

Assigning all transitions from a mission to one partition reduces the risk of related observations from that mission crossing partitions, as can happen with random transition-level splitting. It does not itself guarantee absence of every form of leakage.

15. Running the Optional Examples

From `examples/preprocessing/`, inspect the supported options and run either script with an output directory outside the publication package:

```
python .\prepare_state_only.py --help
python .\prepare_behavior_aware.py --help

python .\prepare_state_only.py --output-root "$env:TEMP\golden_arrows_v3_1_state_only"
python .\prepare_behavior_aware.py --output-root "$env:TEMP\golden_arrows_v3_1_behavior_aware"
```

Both examples accept --source-root, --output-root, and --split-seed. Generated CSVs and supporting outputs are local researcher derivatives, not publication contents.

16. Validation and Quality Control

V3.1 validation checks cover file counts and correspondence, mission distribution, transition and drone cardinality, schema and behavior consistency, numerical integrity, and mission-level structure. The verified results continue on the next page.

16. Validation and Quality Control (continued)

Check	Verified result
Raw / processed files	1,000 raw mission JSON and 1,000 processed JSONL; exact one-to-one filenames
Metadata / mission distribution	1,000 metadata rows; 250 missions per type
Transitions / drone records	113,954 transitions; 3,418,620 derived records (30 per transition)
Schema	golden_arrows_v3_rich_transition_1.0; consistent
Behavior and policy	advanced_behavior_applied = true throughout; documented policy and behavior mode
Numerical integrity	0 NaN; 0 Inf
Mission structure	0 duplicate-tick missions; 0 malformed/structural errors
Transitions per mission	Minimum 9; maximum 373; reported mean 113.95
Completion	1,000 / 1,000 missions completed

These are structural and numerical validation results. They do not establish real-world validity, safety, causal validity, deployment readiness, or operational performance.

17. Validation Artifacts

`validation/dataset_validation_report.txt` and `validation/dataset_statistics.json` document the checks and verified summary values.

18. Intended Research Use

The simulated records may support descriptive analysis of swarm behavior, mission-transition modeling, network-aware coordination, preprocessing methodology, feature-representation studies, and machine-learning methodology research. These are possible uses, not guarantees of model or operational performance.

19. Limitations

This release reflects one documented simulation configuration and four mission types. Its simulation-specific patterns may limit generalization. Further limitations and publication scope are summarized on the next page.

19. Limitations (continued)

The mission controller is scripted, and Advanced Swarm Behavior is scripted telemetry, not a trained policy. Mission transitions and the 30 drone records within each transition are dependent/nested observations. Temporal semantics require task-specific review, and fields under state_after are not automatically pre-action observations. Feature exclusion does not guarantee absence of leakage. The data establish no real-world safety or performance claim, deployment-readiness claim, or causal-validity guarantee.

20. Publication Scope

The publication package contains raw mission records, processed rich transitions, metadata, schemas, validation artifacts, and optional preprocessing examples. It does not contain train.csv, validation.csv, test.csv, generated ML-ready datasets, ml_ready directories, analysis directories, development artifacts, pilot datasets, or staging datasets. Derived ML outputs are generated locally by researchers.

21. Final Package Structure

```
Golden_Arrows_Dataset_V3.1/
|-- data/
|   |-- metadata.csv, raw/missions/, processed/transitions/
|-- schema/
|   `-- mission_schema.json, transition_schema.json, behavior_schema.json
|-- validation/
|   `-- dataset_validation_report.txt, dataset_statistics.json
|-- examples/preprocessing/
|   `-- README.md, prepare_state_only.py, prepare_behavior_aware.py
|-- README.md, LICENSE, CITATION.cff, CHANGELOG.md,
`-- dataset_card.md, dataset_description.pdf
```

22. Related Documentation and Citation

The Golden Arrows Dataset V3.1 package is accompanied by the following documentation and supporting materials:

- **README.md** — Dataset overview, package organization, usage guidance, and reproducibility information.
- **dataset_card.md** — Dataset characteristics, intended uses, limitations, and ethical considerations.
- **CITATION.cff** — Machine-readable citation metadata for the dataset.
- **CHANGELOG.md** — Version history and major changes introduced in V3.1.
- **Schema materials** — Definitions of the dataset structure, fields, and schema version.
- **Validation artifacts** — Structural and quality-control validation results.
- **Preprocessing examples** — Optional state-only and behavior-aware preprocessing workflows.

Dataset DOI

The persistent identifier for Golden Arrows Dataset V3.1 is:

10.5281/zenodo.23030808

This DOI identifies the V3.1 release and should be used when citing or referencing this version of the dataset.

Recommended Citation

Kumar, S. (2026). *Golden Arrows Dataset V3.1: Simulation-Generated Multi-Drone Swarm Transition Dataset*. Zenodo. https://doi.org/10.5281/zenodo.23030808

23. Closing Statement

Golden Arrows Dataset V3.1 is a simulation-generated research resource developed for the analysis of multi-drone swarm transition behavior. The dataset provides structured mission-level and transition-level information while preserving detailed observations of drone state, actions, decisions, network conditions, and Advanced Swarm Behavior.

The V3.1 release contains **1,000 missions and 113,954 transitions**, representing four mission types: PATROL, SURVEILLANCE, SEARCH, and RECOVERY. Each transition contains observations for **30 drones**, resulting in **3,418,620 derived drone-transition records**. These nested drone-transition records are derived from the transition structure and should not be interpreted as independent samples.

The release is accompanied by schema documentation, validation artifacts, reproducibility information, dataset metadata, preprocessing examples, and supporting documentation. The dataset is intended to support research involving swarm behavior analysis, transition modeling, multi-agent learning, behavior-aware machine learning, and related simulation-based studies.

Because the dataset is simulation-generated, its observations should be interpreted within the documented simulation environment, controller configuration, behavioral assumptions, and stated limitations. It is intended as a reproducible research resource and does not represent real-world multi-drone telemetry.

For complete information about dataset structure, usage, preprocessing, validation, reproducibility, and citation, consult the documentation included with the V3.1 release.

GOLDEN ARROWS DATASET V3.1

DISTRIBUTION AND USAGE ADDENDUM

Dated 2026-10-20.

SCOPE OF THIS CORRECTION

This addendum supersedes the distribution wording and file trees in sections 4, 20, and 21; the example commands in section 15; and DOI/publication-status statements in section 22. Earlier directory trees describe the reconstructed layout after extraction. Dataset records and scientific findings are unchanged.

THE EIGHT INDEPENDENT ZENODO DATA ARCHIVES

data_metadata.zip
data_raw_missions_001.zip
data_raw_missions_002.zip
data_raw_missions_003.zip
data_raw_missions_004.zip
data_processed_transitions_001.zip
data_processed_transitions_002.zip
data_processed_transitions_003.zip

Download all eight archives and extract each into the same new, empty destination directory. Each archive preserves the data/ prefix. They are independent ZIP files; no .z01 or .z02 companions are required. Extracting fewer than all eight yields an incomplete dataset.

RECONSTRUCTED DATA LAYOUT

<extraction-directory>/data/metadata.csv
<extraction-directory>/data/raw/missions/ 1,000 JSON files
<extraction-directory>/data/processed/transitions/ 1,000 JSONL files

The temporary failed-upload artifact data/processed/transitions.zip is excluded. Archive verification is separate from dataset validation: the eight archives contain 2,001 intended files with no missing files, duplicate assignments, or unexpected files.

PREPROCESSING EXAMPLES (CORRECTED SECTION 15)

Python 3.9 or newer; standard library only. Extract all eight archives as above. From the release-package root, set --source-root to the extracted data directory and choose output folders outside the package:

```
$source = "C:\path\to\extracted\dataset\data"
python .\examples\preprocessing\prepare_state_only.py --source-root $source --output-root
"$env:TEMP\golden_arrows_v3_1_state_only"
python .\examples\preprocessing\prepare_behavior_aware.py --source-root $source --output-root
"$env:TEMP\golden_arrows_v3_1_behavior_aware"
```

Each example requires exactly 1,000 processed JSONL mission files and validates expected structure/counts before writing local CSV and report outputs. The scripts do not write into the publication package by default and reject output paths inside it.

DOI AND RELEASE DATE (CORRECTED SECTION 22)

CITATION.cff lists DOI 10.5281/zenodo.23030808. The DOI resolver returned HTTP 404 when checked on 2026-10-20; active registration and publication status are unconfirmed. CITATION.cff records release date 2026-10-02, while the changelog describes publication as pending. Both items require maintainer confirmation; neither has been changed by this packaging update.